Day 8-12 Runtime and Performance Report

Project: adaptive-qem-benchmark | Batch date: 2026-11-02

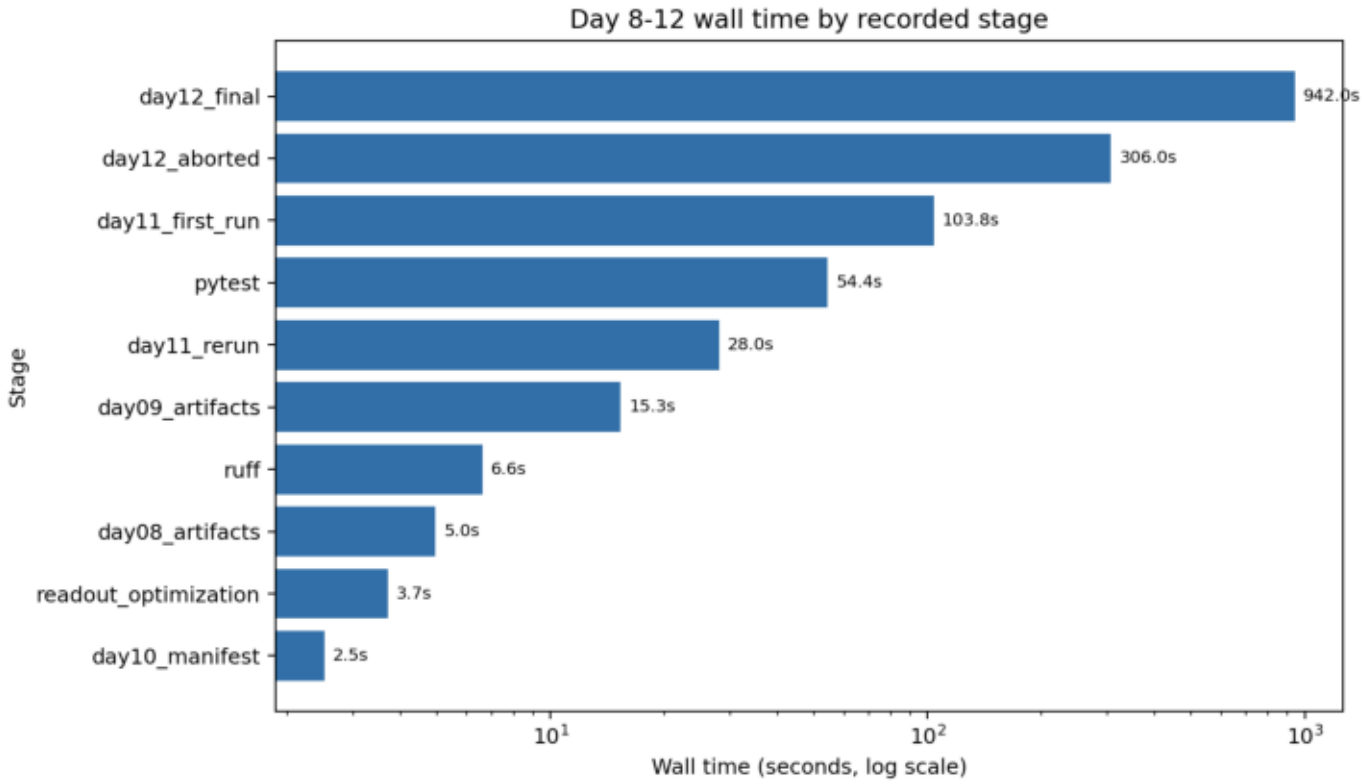
The final 50-repeat GHZ benchmark was the dominant measured stage. Readout calibration optimization cut the comparable Day 11 pilot wall time by about 73%, while retaining fixed shot budgets, deterministic seeds, raw rows, and explicit failure statuses. Times marked estimated use visible session intervals or artifact timestamps rather than unavailable per-stage timers.

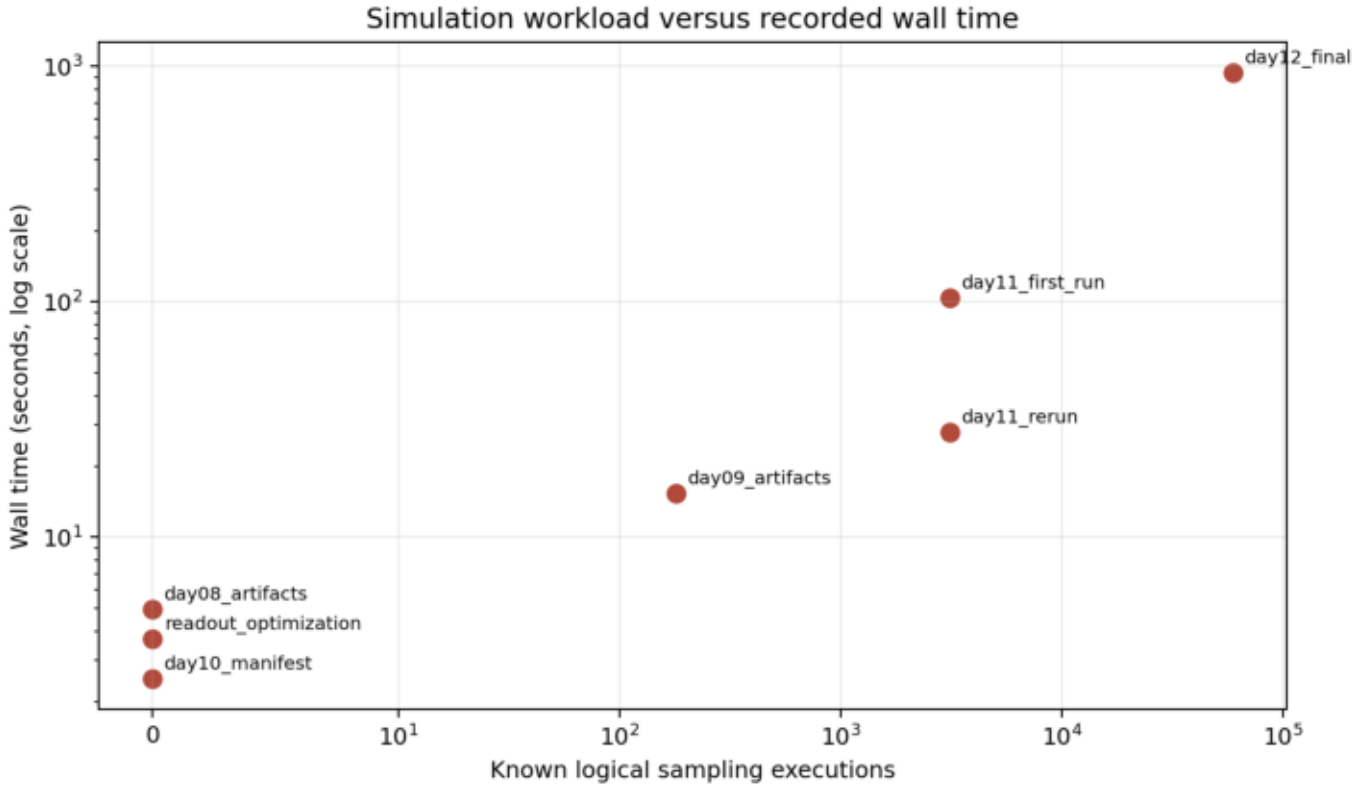
Stage	Day	Seconds	Timing	Cells	Simulations
day08_artifacts	8	4.95	estimated*	9.0	0.0
day09_artifacts	9	15.343	estimated*	60.0	180.0
day10_manifest	10	2.508	estimated*	16200.0	0.0
day11_first_run	11	103.8	measured	240.0	3120.0
readout_optimization	11	3.7	measured	0.0	0.0
day11_rerun	11	28.0	estimated*	240.0	3120.0
day12_aborted	12	306.0	estimated*		
day12_final	12	942.0	measured	5400.0	59400.0
pytest	all	54.37	estimated*		
ruff	all	6.6	estimated*		

Recorded allocation: 95.1% simulation | 0.8% preparation | 4.2% testing/lint

* Estimated durations are identified per row. The aborted Day 12 attempt has unknown completed-cell and simulation counts because no checkpoint was written.

Measured Runtime and Workload





Where time went

Sampling dominated the batch. Linear ZNE executes factors 1, 3, and 5, so each ZNE row requires three circuit samples even though their shots sum to the same budget as competing methods. The frozen GHZ grid then multiplies this work by three widths, four noise levels, three budgets, three methods, and 50 repeats. Deeper folded circuits increase transpilation and simulation cost. Before the optimization, readout calibration launched each basis-state circuit separately; afterward, the same independent assignment channel was sampled with deterministic multinomial draws, sharply reducing process-launch overhead.

Failure context

The final raw table contains 5,372 complete cells and 28 failed cells. Every failure is an explicit ill_conditioned readout status at width 6 and a 1,024-shot total budget: 25 under high noise and 3 under medium noise. These rows remain in the coverage audit and were not clipped, retried, or silently discarded.

What this does not include

No quantum-hardware queue or execution time is included. No LLM or token time is included. Unreported code-writing latency is not reconstructed. The aborted run's completed workload is unknown. Rows labeled estimated use session wait intervals, combined-command timing, or filesystem timestamps and must not be interpreted as instrumented profiler measurements.